

Preface

Friday, August 28, 2020 3:27 PM

1. '

a. Lecturer

Prof. Mustata

b. Comment

C'est difficile

2. '

L1: Basic Concept of Graph

Tuesday, September 1, 2020 10:11 AM

1. Basic Concept of Graph

a. Introduction to Graph

i. Graph Definition

Graph G consists of Vertices $V = V(G)$ and Edge $E = E(G)$ and a map from $e \in E$ to it's end points

ii. Loop Definition

When end points of an edge is equal

iii. Simple Graphs Definition

G no loops; there is at most one edge that is incident to any pair of vertices

iv. Name an Edge

$\{x, y\}$ is the edge (not the set of edge) in incident to x and y vertices

b. Degrees

i. Degree Definition $\deg(x)$

The number of edges incident to x . Attention, loop at x contributes 2 to $\deg(x)$

ii. Neighbor of x $\Gamma(x)$

$\Gamma(x)$ is a set

iii. Theorem: Sum of Degrees

1) Description

For a finite graph G , total sum of degree is twice the number of edge

2) Equation

$$\sum_{x \in V(G)} \deg(x) = 2 \cdot \#E(G)$$

3) Corollary

For a finite graph G , it has an even number of vertices with odd degree

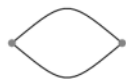
iv. Complete Graph K_n

n vertices, $\binom{n}{2}$ edges

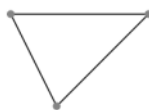
v. Polygon P_n



P_1



P_2



P_3



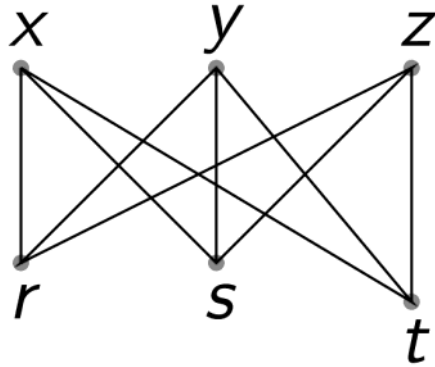
P_4

c. Bipartite $G[A, B]$

i. Definition

$V(G)$ are divided by A and B , while $\forall e \in E(G)$ has one end in A and the other in B

ii. Example



iii. Proposition (Proof by Counting Vertices with Edges)

$G[A, B]$ such that $\forall_{x \in A, y \in B, \{x, y\} \in E(G)} \deg x \geq \deg y$ and have no isolated point

$\Downarrow$

$\#A \leq \#B$, equality iff $\forall_{x \in A, y \in B, \{x, y\} \in E(G)} \deg x = \deg y$

d. Isomorphism: an identification

i. Definition

Given two graphs G_1, G_2 , an isomorphism from G_1 to G_2 is given by two **bijective** maps $\varphi: V(G_1) \rightarrow V(G_2), \psi: E(G_1) \rightarrow E(G_2)$, so that e joins a and b iff $\psi(e)$ joins $\varphi(a)$ and $\varphi(b)$

ii. Automorphism: symmetry

Two vertices of a graph are similar if there's an automorphism maps one point to the other one

e. Subgraph

i. Subgraph

1) Definition

Definition. Given a graph G , a **subgraph** H of G is a graph H such that

i) $V(H) \subseteq V(G)$.

ii) $E(H) \subseteq E(G)$.

iii) If $e \in E(H)$, then its ends in H are the same as its ends in G .

2) **Question**

Does ii) suggest iii)?

ii. Spanning Subgraph

1) Definition

$V(H) = V(G)$

2) Comment

We can create a spanning subgraph by removing edges

iii. Span by

1) Definition

Given a graph G and a subset S of $V(G)$, then the subgraph **spanned by** S is the subgraph of G whose

- set of vertices is S and

- set of edges consists of those edges in G that join vertices in S .

2) Comment

Like linear algebra, "span by" implicitly suggest the concept of subgraph

f. A Walk

i. Definition: A Sequence

$$x_0, e_1, x_1, e_2, \dots, e_n, x_n$$

ii. Notation

$x_0, e_1, x_1, e_2, \dots, e_n, x_n$ is a $x_0 - x_n$ walk of length n

iii. More Definitions

1) A Path

Edge of walk are distinct

2) Closed

$$x_0 = x_n$$

3) Simple

Vertices are distinct (automatically a path!)

4) Simple closed

$x_0 = x_n$, while $x_0, \dots, x_{n-1}$ are distinct (this is not simple!)

g. Connectedness

i. We can Go backward

ii. We can concatenation two walks

iii. Connected

If any two points have walk

iv. Convention

Empty graph is not connected

L2: Eulerian and Hamiltonian

Thursday, September 3, 2020 10:08 AM

1. '

a. Adjacency Matrix A_G

i. Definition

$$a_{i,j} = \#\{x_i, x_j\}$$

ii. Comment

- 1) Fully determine isomorphism
- 2) Symmetric matrix
- 3) Simple graph have 0 diagonals

b. d – regular Graph

i. Definition

Every vertex x is of degree d

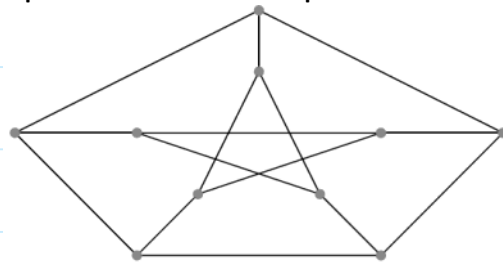
ii. Example: Complete Graph

K_n is regular of degree $n - 1$

iii. Example: Polygon

2

iv. Example: Petersen Graph



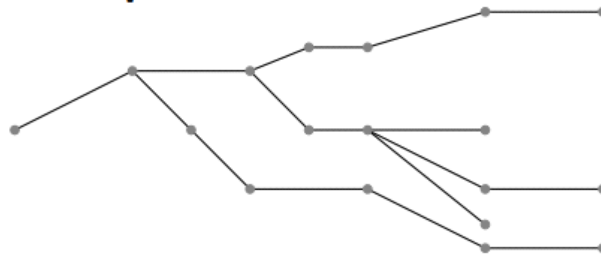
c. Tree

i. Definition

Contains no polygons as subgraphs

ii. Example

Example of a tree.



d. Eulerian Circuit and Eulerian Graph

i. Definition

Closed path going through all edges exactly once

ii. Introduction

7 bridge problem

iii. Theorem

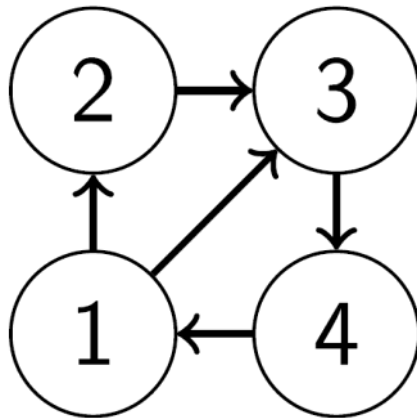
Finite graph G with no isolated vertices, Eulerian **iff** G is connect and very vertex have even degree

e. Directed Graph

i. Definition

Edges are of ordered pair of vertices

ii. Example



iii. Strong Walk

Walk in the direction

iv. Strongly Connected

Every two vertices have strong walk in between

v. Theorem: Equivalence of Directed Eulerian Circuit

Strongly connected and in-degree = out-degree

f. Hamiltonian Circuit

i. Definition

Simple-closed path pass through every vertices

ii. Remark

A Hamiltonian circuit is the same as giving a spanning polygon for G

iii. Comparison: Eulerian and Hamiltonian

They don't imply each other

Hamiltonian doesn't have to pass through all edges

iv. Ore Theorem: A Sufficient Condition

1) Theorem

Theorem 3 (Ore). If G is a simple graph with $n \geq 3$ vertices such that for any two non-adjacent vertices x and y , we have

$$\deg(x) + \deg(y) \geq n,$$

then G is Hamiltonian.

2. \'

L3: Tree (I)

Tuesday, September 8, 2020

10:17 AM

1. Trees

a. Introduction

i. Definition

Connected graph without subgraph isomorphic to a polygon P_n

ii. Three Equivalency

G is a tree (Connected graph without subgraph isomorphic to a polygon P_n)

$\Leftrightarrow$

$\forall_{x \neq y \in V(G)} \exists \text{ unique path from } x, y$

$\Leftrightarrow$

G is connected and contains no closed path (of positive length)

b. Properties

i. Lemma 1: Existence of $\deg(x) = 1$

If G is a tree with $\#V(G) > 1$, then there is $x \in V(G)$ with $\deg(x) = 1$

ii. Lemma 2 and 3: Conservation

1) Lemma 2

A connected graph G remove a $\deg(1)$ vertex will be still connected

2) Corollary

A tree G remove a $\deg(1)$ vertex will be still a tree

3) Lemma 3

A tree G connect a new $\deg(1)$ vertex will be still a tree

iii. Proposition 2

1) Statement

If is a connected graph with n vertices and d edges, then $d \geq n - 1$, and **equality holds iff G is a tree**

2) Proof

By induction over n

c. Cayley's Formula

i. Formula

The number of trees on a set of $n \geq 2$ vertices is n^{n-2}

ii. Multinomial

$$\binom{n}{m_1, \dots, m_k} = \frac{n!}{m_1! \dots m_k!} = \sum_{i=1}^k \binom{n-1}{m_1, \dots, m_i-1, m, \dots, m_k}$$

d. '

L4: Tree (II)

Thursday, September 10, 2020 10:03 AM

1. General Tree

a. Previously

i. Three Equivalency

G is a tree (Connected graph without subgraph isomorphic to a polygon P_n)

$\Updownarrow$

$\forall_{x \neq y \in V(G)} \exists \text{ unique simple path from } x, y$

$\Updownarrow$

G is connected and contains no closed path (of positive length)

ii. Convention

All path are of positive length

iii. Number Statement

If G is a finitely connected graph, then $\#E(G) \geq \#V(G) - 1$, and **equality holds iff G is a tree**

iv. Cayley's Formula

The number of trees on a set of $n \geq 2$ vertices is n^{n-2}

b. Cayley's Formula

i. Multinomial

$$\binom{n}{m_1, \dots, m_k} = \frac{n!}{m_1! \dots m_k!} = \sum_{i=1}^k \binom{n-1}{m_1, \dots, m_i-1, m_{i+1}, \dots, m_k}$$

ii. 1st Proof

$$t(n; d_1, \dots, d_n) = \binom{n-2}{d_1-1, \dots, d_n-1}$$

iii. 2nd Proof

Pruffer's construction

2. Weighted Graphs

a. Introduction

i. Definition

$c: E(G) \rightarrow R$

ii. Problem

We want a spanning tree with minimal $c(T)$

b. Forest

i. Definition

All connected component of finite graph G are trees

ii. Further Number Statement

If G is a finite graph, then $\#E(G) \geq \#V(G) - \#\{\text{connected components of } G\}$, and **equality holds iff G is a forest**

iii. Notation

$\forall_{S \subseteq E(G)} (G: S)$ indicates the spanning subgraph of G whose edge set is S

iv. Definition of Independent

For finite graph G , $S \subseteq E(G)$ is independent if $(G: S)$ is a forest

c. The Greedy Algorithm

i. Algorithm

Algorithm. Let G be a finite, connected, weighted graph. Let $n = \#V(G)$.

Step 1. Choose $e_1 \in E(G)$ such that $c(e_1)$ is minimal. Note: the graph $(G : \{e_1\})$ is a forest, with $n - 1$ connected components.

Step $i \leq n - 1$. Suppose that we have chosen $e_1, \dots, e_{i-1} \in E(G)$ and we know that $(G : \{e_1, \dots, e_{i-1}\})$ is a forest, with $n - i + 1$ connected components.

Step i , cont'd. We choose $e_i \in E(G)$ with ends in two different components of $(G : \{e_1, \dots, e_{i-1}\})$ and such that $c(e_i)$ is minimal among such possibilities.

ii. Claim of Optimization

1) Statement

$\forall$ spanning tree of T' of G $c(T) \leq c(T')$, where T is produced by greedy algorithm

2) Proof

Claim'. For every $1 \leq i \leq n - 1$, we have $c(e_i) \leq c(a_i)$.

We argue by contradiction. Suppose that $i \leq n - 1$ is such that $c(e_i) > c(a_i)$. Note that we also have

$$c(a_i) \geq c(a_{i-1}) \geq \dots \geq c(a_1).$$

Since $c(e_i)$ is strictly larger than all $c(a_j)$, with $j \leq i$, we deduce from the fact that at Step i none of the edges $a_1, \dots, a_i$ was chosen that each of $a_1, \dots, a_i$ had both ends in the same connected component of $(G : \{e_1, \dots, e_{i-1}\})$.

3. Depth-first Search Tree

a. Root

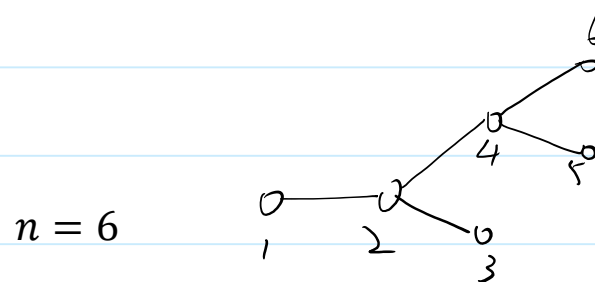
b. Ancestors and descendent

4. '

Question on Pruffer's Proof of Cayley's Formula

Thursday, September 10, 2020

2:12 PM



i	x_i	y_i	T_i
1	1	2	<
2	3	2	>
3	2	4	<
4	5	4	>
5	4	6	

L5: Depth-First Search Tree

Tuesday, September 15, 2020

10:02 AM

1. Depth-First Search Tree

a. Construction

Description of the procedure.

Step 0. Pick an arbitrary vertex v_0 of G . We consider the tree T_0 with vertex v_0 and no edges.

Step $k + 1$. Suppose that we have already picked distinct vertices $v_0, \dots, v_k$ of G and a subgraph T_k of G on these vertices, which is a tree.

Step $k + 1$, cont'd. If $k + 1 < \#V(G)$, since G is connected, there is $\ell \leq k$ such that v_ℓ is adjacent to a vertex of G which is not a vertex of T_k . We choose ℓ maximal with this property and one such edge, whose end different from v_ℓ is denoted v_{k+1} (note: while ℓ is uniquely determined by maximality, both v_{k+1} and the edge are not unique).

We let T_{k+1} be the graph obtained from T_k by adding v_{k+1} and this edge. We have already seen (Lemma 3 from Lecture 3) that in this case T_{k+1} is a tree as well.

On the other hand, if $k + 1 = \#V(G)$, then the procedure stop. In this case T_k is a spanning tree of G .

Conclusion. We end with a numbering of the vertices of G and a spanning tree of G .

b. Root and Orientation

i. Root

A choice of x_0 in tree T

ii. Definition: Ancestors & Descendent

Iff $y \in V(G)$ appears on a unique path from x_0 to x , then y is an ancestors of x

iii. Convention

x is both an ancestor and descend ot itself

iv. Parent

The closest, if any, ancestor

v. Remark

Two vertices if adjacent in T , then one is the parent of the other

c. Proposition 1

i. Statement

If x, y adjacent in G , then x is either ancestor or descendent of y in T

ii. Proof

$x = v_k, y = v_{k'}$. By construction, they follows the statement

d. Proposition 2

i. Definition of Isthmus / Bridge

A bridge is an edge, who's deletion results in a disconnect graph

ii. Statement

For a depth-first search tree T of G with root v_0 , if x is the parent of y in T , and $\{x, y\}$ not an isthmus in G , then there is an edge $e \in E(G), \notin E(T)$ joining a descendent a of y with ancestor b of x

e. Theorem

i. Statement

If G is a finite connected graph **without isthmuses**, then G admites an orientation that makes it a strongly connected digraph.

ii. Proof

2. Color

a. Basics

i. Definition of Coloring

$V(G) \rightarrow C$

ii. Proper Coloring

Ends of every edge have different color

iii. Notation: $k - colored$

$\#C = k$

iv. Convention

No loop, simple graph

v. Chromatic Number

$\chi(G)$ minimum posible k-color

b. Examples

$\chi(G)$		G
$= 1$	$\Leftrightarrow$	$E(G) = \emptyset$
≤ 2	$\Leftrightarrow$	Bipartite
$= n$	$\Leftrightarrow$	K_n
$= 2, 3$	$\Leftarrow$	P_n
$= 2$	$\Leftarrow$	T

c. Brooks' Theorem

i. Statement

3. ' _____

L6: Colored Map: Brook's Theorem

Thursday, September 17, 2020

10:01 AM

1. Colored Map

a. Brook's Theorem

i. Statement

$$\left(\begin{array}{l} d \geq 3, G \text{ a finite graph without loop} \\ K_{d+1} \text{ is not isomorphic with a subgraph of } G \\ \text{All vertices of } G \text{ has degree } \leq d \end{array} \right) \Rightarrow \chi(G) \leq d$$

ii. Proof

1) Notation: Maximum Degree

$$\Delta G := \max_{x \in V(G)} \deg(x)$$

I cannot understand the proof

2) Step 1: Assume Counterexample with Minimum Vertices

G is the minimum vertice counterexample.

3) Step 2: Constructed Subgraph H has Chromatic Number d

H is construct by deleting an arbitrary chosen x from G

Since G is the minimal counterexample, $\chi(H) \leq d$, however, we must take equality because otherwise we can use the unassigned color to form $\chi(G) \leq d$

4) Step 3: $\Gamma(x) = \{x_1, \dots, x_l\}$, adjacent of x must be d -colored

Otherwise, we can use the unassigned color for x . We can thus name the colors with $1, \dots, d$

5) Step 4: Any x_i and x_j are connected in Induced Subgraph H_{ij}

a) Definition of H_{ij}

H_{ij} is the subgraph whose color in H are i and j .

b) x_i and x_j are in Connected Component C_{ij}

Suppose x_i and x_j not connected in H_{ij} , then we can switch color of x_i so that x_i has color of j , but this will lead to contradiction with Step 3. Thus x_i and x_j are in a connected component of H_{ij} . Name the that component C_{ij}

6) Step 5: C_{ij} is a Simple Path

a) x_i is an end

$\deg_{H_{ij}}(x_i) = 1$, because otherwise $\Gamma(x_i)$ will be at most $d - 2$ colored. Then we will can recolor x_i , but this will contridict

Step 3.

b) All Intermediate Point y has $\deg_{H_{ij}} y = 2$

If C_{ij} contains any vertex y : $\deg_{H_{ij}} y \geq 3$, then $\chi(\Gamma_H(y)) \leq d - 2$ thus we can recolor y , but that will disconnect C_{ij} , and will contradict Step 4

7) Step 6: C_{ij} does not intersect in intermediate point z

Suppose $z \neq x_i$ and $z \in C_{ij} \cap C_{ik}$, then z must be of i color, and will have two neighbors with j color and two neighbors with k color. Thus, $\chi(\Gamma_H(z)) \leq d - 2$, and we thus can recolor with a color $\notin \{i, j, k\}$, but this will disconnect C_{ij} and C_{ik} , thus contradicting step 5.

8) Step 7: $\sim K_{d+1} \subseteq G$

a) Preparation

x_1 and x_2 are not connected by an edge, otherwise $K_{d+1} \subseteq G$. Thus, there exists $a \in C_{12}$. Switch color 1 and 3 for C_{13} . Denote new path connecting i and j as C'_{ij}

b) $a \in C'_{23}$

Since a is the only color 2 neighbor of x_1 , which is now color 3, $a \in C'_{23}$

c) $a \in C'_{12}$

According to Step 5, since x_1 is the only vertex that change color in C_{23} , a must be within C'_{12}

d) Contradiction

b) and c) show that $C'_{12} \cap C'_{23} \neq \{x_2\}$. Thus, our assumption of the existence of G is wrong.

L7: Kneser's Graph

Tuesday, September 22, 2020 10:54 AM

1. Coloring

a. Kneser's Graph Definition

$KG_{n,k}$ is a graph of vertex set $\binom{[n]}{k}$, where A and B are joined by an edge iff $A \cap B = \emptyset$

b. Kneser's Bound for $\chi(KG_{n,k})$

i. Statement

$$\chi(KG_{n,k}) \leq n - 2k + 2$$

ii. One Such Coloring

$$c(A) = \min\{\min(x|x \in A), n - 2k + 2\}$$

c. Kneser's Conjecture

i. Conjecture

$$\chi(KG_{n,k}) = n - 2k + 2$$

ii. Supporting Theorem:

1) Borsuk-Ulam Theorem

If $f: S^n \rightarrow \mathbb{R}^n$ is a continuous function, then there is $x \in S^n$ such that $f(x) = f(-x)$.

2) Definition of n-dimension Sphere

$$S^n := \{(a_0, \dots, a_n) \in \mathbb{R}^{n+1} | a_0^2 + \dots + a_n^2 = 1\}$$

iii. Proof

1) Corollary

If $A_0, \dots, A_n \subseteq S^n$ are open or closed subsets such that $S^n = A_0 \cup \dots \cup A_n$, then there is $x \in S^n$ and $0 \leq i \leq n$ such that $x, -x \in A_i$

iv. More Reading

<https://www.sciencedirect.com/science/article/pii/0097316578900237>

d. Ramsey Theory

i. Theorem 1

Theorem 1. Suppose that the edges of K_n are colored red or blue and $r_1, \dots, r_n$ are such that

r_i = number of red edges with vertex i as endpoint.

If Δ is the number of monochromatic triangles, then

$$\Delta = \binom{n}{3} - \frac{1}{2} \cdot \sum_{i=1}^n r_i(n-1-r_i).$$

ii. Corollary

1) Notation: Round Down

$$\lfloor 1.1 \rfloor = 1$$

2) Statement

$$\Delta \geq \binom{n}{3} - \left\lfloor \frac{n}{2} \left\lfloor \frac{(n-1)^2}{4} \right\rfloor \right\rfloor$$

3) Proof: Arithmetic vs Geometric Mean

$$r_i(n-1-r_i) \leq \frac{(n-1)^2}{4}$$

iii. Theorem 2

Theorem 2. Given $s, t \geq 2$, there is a number $R(s, t)$ such that for every graph on $n \geq R(s, t)$ vertices, there is either a set of s vertices, no 2 of them adjacent, or there is a set of t vertices, any two of them adjacent.

Example. We saw that we can take $R(3, 3) = 6$.

2. '

L8: Ramsey's Theorem

Thursday, September 24, 2020 10:05 AM

1. Ramsey's Theorem

a. Introduction

"There can't be complete chaos". If a structure is large enough, no matter how complicated, there is a smaller subsystem that exhibits a special structure

b. Theorem 2:

i. Statement

Given $s, t \geq 2$, there is a number $R(s, t)$ such that for every graph on $n \geq R(s, t)$ vertices, there is either a set of s vertices, no 2 of them adjacent, or there is a set of t vertices, ant two of them adjacent.

ii. Example: $R(3,3) = 6$

Any party with at least 6 people will contains a group of three mutual friends or a group pf three mutual non-friends

c. Theorem 3: Ramsey's Theorem

i. Statement

Given $r \geq 1$ and $q_1, \dots, q_s \geq r$ there is a number N such that the following holds: given a set S with $n \geq N$ elements such that all $\binom{n}{r}$ r -subsets of S are divided into s mutually exclusive families $T_1, \dots, T_s$ ("colors"), there is i with $1 \leq i \leq s$ and some q_i -subset of S for which every r -subset is in T_i

ii. Remark

For $r = 2$ (r -subset now describe edge) and $s = 2$ (two color), we have theorem 2

iii. Notation

$$N \geq N(q_1, \dots, q_s; r)$$

d. Theorem 4: Erdos-Szekeres

Theorem 4 (Erdős-Szekeres). For every $n \geq 1$, there is $N(n)$ such that any collection of $N \geq N(n)$ points in the plane, with no 3 of them on a line, has a subset of n points forming a convex polygon.

Proof. We will use the following fact: given n points in the plane, with no 3 of them on a line, they form a convex polygon if and only if any 4 of them form a polygon.

e. Estimating Ramsey's Number

i. "Obviously"

1) Obvious 1

$$N(p, q; 1) = p + q - 1$$

2) Obvious 2

$$N(p, r; r) = p$$

ii. Formula 1

1) Inequality

$$N(p, q; r) \leq N(N(p-1, q; r), N(p, q-1; r); r-1) + 1$$

2) Corollary

$$N(p, q; 2) \leq \binom{p+q-2}{p-1}$$

L9: Dependency Graph

Tuesday, September 29, 2020

10:18 AM

1. Ramsey's Number

a. Previously: Estimating Ramsey's Number

i. "Obviously"

1) Obvious 1

$$N(p, q; 1) = p + q - 1$$

2) Obvious 2

$$N(p, r; r) = p$$

ii. Formula 1

1) Inequality

$$N(p, q; r) \leq N(N(p-1, q; r), N(p, q-1; r); r-1) + 1$$

2) Corollary

$$N(p, q; 2) \leq \binom{p+q-2}{p-1}$$

b. Lower Bound

i. Theorem

For $p \geq 2$, we have $N(p, p; 2) \geq 2^{\lfloor p/2 \rfloor}$, with strict inequality for $p \geq 3$

ii. Proof

For K_n , total possible coloring is $2^{\binom{n}{2}}$, coloring such that K_p is monochromatic is $2^{\binom{n}{2} - \binom{p}{2} + 1}$. The former is larger or equal to the latter

c. Upper Bound

i. Theorem

$$N(p, p; 2) \leq 4^p$$

ii. Proof

[Formula 1](#)

d. Probability Method

i. Introduction

1) Probability Space X , Events $A_1, \dots, A_n$

$$\Pr[A_i] = \frac{\#A_i}{\#X}, \quad \Pr[\overline{A_i}] = 1 - \Pr[A_i]$$

ii. Independency

1) Definition

$$A, B \text{ independent} \Leftrightarrow \Pr[A] = \Pr[A|B]$$

2) Lemma 1

If A and B are independent, then A and $\bar{B}$ are independent too

iii. Dependency Graph

1) Definition

Given events $A_1, \dots, A_n$, a **dependency graph** for these events is a graph G on the set $\{1, \dots, n\}$ with the property that for every i and every $j_1, \dots, j_r$, with all j_k different from i and non-adjacent to i , the events $\overline{A_i}$ and $\overline{A_{j_1}} \cap \dots \cap \overline{A_{j_r}}$ are independent.

2) Remark

When two events are dependent, the corresponding vertices are adjacent

iv. A Stronger Lower Bound

There is $c > 0$ such that $N(p, p; 2) \geq cp2^{p/2}$ for $p \gg 0$

e. '

f. '

L10: Hall's Marriage Theorem

Thursday, October 1, 2020

10:00 AM

1. Hall's Marriage Theorem

a. Introduction

i. Bipartite Graph

ii. Convention

We assume simple graph

iii. Matching

At most one edge.

iv. Complete Matching

A complete matching from X to Y is a injective map.

This however doesn't guarantee all Y element has matching

b. When We have Complete Matching

1) Hall's Marriage Theorem

A complete matching Λ from X to Y exists **iff** for any $A \subseteq X$, $\#\Gamma(A) \geq \#A$

i. Proof of Sufficiency

Recursive construction

c. System of Distinct Representatives

i. SDR

Setup. Suppose that S is a finite set and $A_0, \dots, A_{n-1}$ are subsets of S . A **system of distinct representatives** (or **SDR**, for short) is given by n distinct elements $a_0, \dots, a_{n-1} \in S$ such that $a_i \in A_i$ for $0 \leq i \leq n-1$.

ii. Property (H)

For sets $A_0, \dots, A_{n-1}$, $\forall k$ the union of k sets has $\geq k$ elements

iii. Critical Block

A set of union of k that has precisely k elements

iv. SDS Theorem

$A_0, \dots, A_{n-1}$ satisfies (H) **iff** they have **SDR**

d. Application

i. König's Theorem

Theorem 2 (König). Let $A = (a_{ij})$ be an $m \times n$ matrix, with $a_{ij} \in \{0, 1\}$ for all i and j . In this case the minimum number m of lines of A that contain all 1's is equal to the maximum number M of 1's, no two on the same line.

In the above: a **line** of A is either a row or a column of A .

ii. Birkhoff Theorem

Theorem 3 (Birkhoff). Let $A = (a_{ij})$ be an $n \times n$ matrix, with all $a_{ij} \in \mathbb{Z}_{\geq 0}$. If A satisfies

- i) For every j , we have $\sum_i a_{ij} = \ell$, and
 - ii) For every i , we have $\sum_j a_{ij} = \ell$,
- then A is a sum of ℓ permutation matrices.

e. A Lower Bound for numbers of SDR

i. Definitions

Notation. Given integers $m_0 \leq m_1 \leq \dots \leq m_{n-1}$, we put

$$F_n(m_0, \dots, m_{n-1}) := \prod_{i=0}^{n-1} \max\{1, m_i - i\}.$$

Given arbitrary integers $a_0, \dots, a_{n-1}$, we put

$$f_n(a_0, \dots, a_{n-1}) := F_n(m_0, \dots, m_{n-1}),$$

where $(m_0, \dots, m_{n-1})$ is a nondecreasing rearrangement of $(a_0, \dots, a_{n-1})$.

ii. Lemma

$f_n: \mathbb{Z}^n \rightarrow \mathbb{Z}_{>0}$ is nondecreasing w.r.t each variable

iii. Lower Bound

Theorem 4. Given the subsets $A_0, \dots, A_{n-1}$ of the finite set S , with $\#A_i = m_i$, if the sets satisfy property (H), then the number $N(A_0, \dots, A_{n-1})$ of SDRs for $A_0, \dots, A_{n-1}$ satisfies.

$$N(A_0, \dots, A_{n-1}) \geq f_n(m_0, \dots, m_{n-1}).$$

In particular, there is an SDR for $A_0, \dots, A_{n-1}$.

L11: Planar Graph

Monday, October 12, 2020 10:11 PM

1. Planar Graph

a. Basics

i. Definition: Planar Graph

A finite graph G is planar if it has some planar drawing

ii. Definition: Subdivision

"dividing an edge into two edges"

iii. Remark

If H is a subdivision of G , then G is planar iff H is planar

b. Jordan Curve Theorem

i. Definition: Jordan Curve

A simple closed curve in $\mathbb{R}^2$

ii. Jordan Curve Theorem

For every Jordan curve, its complement in $\mathbb{R}^2$ has precisely two connected components, one bounded and one unbounded

c. Euler's Formula

i. Euler's Theorem

For every planar graph G , $\#F - \#E + \#V = 2$

ii. Corollary 1:

1) Statement

If G is a connected planar simple graph, with $n \geq 3$ vertices, then $\#E(G) \leq 3n - 6$

2) Key Step

$$\#F \leq \frac{2}{3} \#E(G)$$

iii. Corollary 2:

Let G be a connected planar simple graph. If $g \geq 3$ is such that $\#E(G) \geq g$ and G has no P_n with $n < g$, then $\#E(G) \leq \frac{g}{g-2} (\#V(G) - 2)$

iv. Theorem 2

A graph G is not planar iff it contains a subgraph which is a subdivision of either K_5 or $K_{3,3}$

d. Four Color Theorem

i. Theorem

If G is a planar graph, then $\chi(G) \leq 4$

L12: Extremal Graph

Tuesday, October 13, 2020 11:46 AM

1. '

a. Introduction

Given a graph G on n vertices, how big certain global invariants can be when G does not exhibit certain local substructures? What are the graphs without such local substructures that have maximal global invariants?

b. Graph without Triangle

i. Bipartite Graph Proposition

If G is a bipartite graph on n vertices, then G can have $\leq n^2/4$ edges

ii. No Triangle Proposition

If G is a graph on n vertices that does not contain any triangle, then $\#E(G) \leq n^2/4$

c. Graph without K_{t+1}

i. How to Construct Such Graph

Let $V = V_1 \cup V_2 \cup \dots \cup V_t$ and then we only allow edges from vertices in two different set

ii. Maximum Number of Edge within Constructed Graph

$$\#E(G) = \sum_{i < j} \#(V_i) \cdot \#(V_j)$$

iii. Requirement for Reaching Maximal when $n = \sum \#(V_i)$ is fixed

$$\#(V_i) - \#(V_j) \in \{0, -1, 1\}$$

d. Turan Graph $T_{n,t}$

i. Definition: Turan Graph $T_{n,t}$

if $n = tk + r$, with $r \in \{0, 1, \dots, t-1\}$, then after reordering the V_i , we may assume that $d_i = k+1$ for $1 \leq i \leq r$ and $d_i = k$ for $r+1 \leq i \leq t$.

The corresponding graph is the **Turán graph** $T_{n,t}$.

ii. Turan Theorem

Theorem (Turán). For every $t \geq 2$, among all graphs on n vertices that contain no K_{t+1} , the graph $T_{n,t}$ has the most edges; moreover, it is unique with this property.

iii. Number of Edge of Turan Graph

$$\frac{n^2}{2} \left(1 - \frac{1}{t} \right) + O(n)$$

e. Erdos-Stone Theorem

i. Definition

$ex(n, H)$ the maximum number of edges of a graph on n vertices that does not contain any graphs isomorphic to H

ii. Erdos-Stone Theorem

$$\forall H: E(H) \neq \emptyset \quad \forall \epsilon > 0, n \gg 0 \quad \frac{1}{2} \left(1 - \frac{1}{\chi(H) - 1} - \epsilon \right) n^2 < ex(n, H) < \frac{1}{2} \left(1 - \frac{1}{\chi(H) - 1} + \epsilon \right) n^2$$

iii. Proof of Lower Bound

Defined k such that $\chi(H) = k + 1$. We already know that $\chi(T_{n,k}) = k$. Thus

$$ex(n, H) \geq \#E(T_{n,k})$$

iv. Remark

$$\text{For } \chi(H) \geq 3, \lim_{n \rightarrow \infty} \frac{\text{ex}(n, H)}{n^2} = \frac{1}{2} \left(1 - \frac{1}{\chi(H)-1} \right) > 0$$

f. Examples

i. Polygon 4

1) Result

$$\text{ex}(n, P_4) \leq \frac{n}{4} (1 + \sqrt{4n-3})$$

2) Proof: Two Ways of Counting

Proof. For every $v \in G$, let's put $d_v = \deg(v)$. We consider

$$F = \{(u, v, w) \mid \{u, v\} \in E(G), \{u, w\} \in E(G), v \neq w\}.$$

We count $\#F$ in two ways:

- Every $u \in V(G)$ contributes $d_u(d_u - 1)$ to F , hence $\#F = \sum_{u \in V(G)} d_u(d_u - 1)$.
- For every pair of distinct vertices (v, w) , there is at most one $(u, v, w) \in F$ since G contains no P_4 , hence $\#F \leq n(n-1)$.

ii. Theorem 2

1) Theorem

Theorem 2. For every $t \geq 2$, there is $c > 0$ such that

$$\text{ex}(n, K_{t,t}) < c \cdot n^{2-\frac{1}{t}} \quad \text{for all } n \gg 0.$$

2) Key Argument

the set consisting of the neighbors of i has precisely $\binom{d_i}{t}$ subsets with t elements. Since G contains no $K_{t,t}$, when we vary i , every such subset appears $\leq (t-1)$ times. We thus have

$$\sum_{i=1}^n \binom{d_i}{t} \leq (t-1) \binom{n}{t} \leq (t-1) \frac{n^t}{t!}. \quad (2)$$

iii. Theorem 3

Theorem 3. There is $c > 0$ such that

$$\text{ex}(P_{2k}, n) \leq c \cdot n^{1+\frac{1}{k}} \quad \text{for all } n.$$

g.

h. '

i. '

L13: Szemerédi Regularity Lemma

Friday, October 16, 2020

2:36 PM

1. Edge Density

a. Definition

Definition. Given a graph G with vertex set V and two (nonempty) subsets $X, Y \subseteq V(G)$, we put

$$E_G(X, Y) = \{(x, y) \in X \times Y \mid \{x, y\} \in E(G)\}$$

and the edge density between X and Y is given by

$$d_G(X, Y) = \frac{|E_G(X, Y)|}{|X| \cdot |Y|} \in [0, 1].$$

b. ϵ -regular

i. Definition

Definition. A pair (X, Y) of nonempty subsets of V is ϵ -regular if for every subsets $A \subseteq X$ and $B \subseteq Y$, with $|A| \geq \epsilon|X|$ and $|B| \geq \epsilon|Y|$, we have

$$|d_G(A, B) - d_G(X, Y)| \leq \epsilon.$$

ii. Comment

0-regular is most demanding

c. ϵ -partition

i. Definition

Definition. A partition $\mathcal{P}$ of V given by $\{V_1, \dots, V_r\}$ is an ϵ -regular partition if

$$\sum_{\substack{1 \leq i, j \leq r \\ (V_i, V_j) \neq \epsilon\text{-regular}}} |V_i| \cdot |V_j| \leq \epsilon |V|^2.$$

ii. Comment

? Why we count $(V_i, V_j) \neq \epsilon\text{-regular}$ instead of $(V_i, V_j) = \epsilon\text{-regular}$

d. Theorem

Theorem 4 (Szemerédi's Regularity Lemma). For every $\epsilon > 0$ and every positive integer m , there is an integer $M \geq m$ such that for every graph G on a set of vertices V , with $|V| \geq M$, there is an ϵ -regular partition of V into k sets, with $m \leq k \leq M$.

2. '

L14: Hermitian Matrix

Tuesday, October 20, 2020 10:19 AM

1. '

a. Hermitian Matrices

i. Hermitian Matrices

1) Definition

Definition. A square complex matrix $A = (a_{ij}) \in M_n(\mathbf{C})$ is Hermitian if $A^t = \bar{A}$; explicitly, we have

$$a_{j,i} = \overline{a_{ij}} \quad \text{for all } 1 \leq i, j \leq n.$$

2) In Real Domain

Note: if $A \in M_n(\mathbf{R})$, then $\bar{A} = A$, hence A is Hermitian if and only if it is symmetric.

ii. Inner Product

1) Definition

Recall that $\mathbf{C}^n$ carries a standard inner product $\langle -, - \rangle$ given by

$$\langle u, v \rangle = \sum_{i=1}^n u_i \bar{v}_i \quad \text{for } u = (u_1, \dots, u_n), v = (v_1, \dots, v_n).$$

Key property: for all u , we have $\langle u, u \rangle \geq 0$, with equality if and only if $u = 0$.

2) Hermitian and Inner Product

If A is Hermitian, then it follows from the definition and the above formula that $\langle Au, v \rangle = \overline{\langle Av, u \rangle}$. In particular, we have $\langle Au, u \rangle \in \mathbf{R}$ for all u .

iii. Min-Max Theorem

1) Convention

$$\lambda_1 \leq \dots \leq \lambda_n$$

2) Rayleigh Quotient

We consider the Rayleigh quotient

$$R_A(u) := \frac{\langle Au, u \rangle}{\langle u, u \rangle} \in \mathbf{R} \quad \text{for } u \in \mathbf{C}^n \setminus \{0\}.$$

Clear: $R_A(u) = R_A(\alpha u)$ for all $\alpha \in \mathbf{C}$, $\alpha \neq 0$.

3) Min-Max Theorem

Theorem 1. For every k , with $1 \leq k \leq n$, we have

$$\begin{aligned} \lambda_k &= \min \left\{ \max \{ R_A(u) \mid u \in U, u \neq 0 \} \mid U \subseteq \mathbf{C}^n, \dim(U) = k \right\} \\ &= \max \left\{ \min \{ R_A(u) \mid u \in V, u \neq 0 \} \mid V \subseteq \mathbf{C}^n, \dim(V) = n - k + 1 \right\}. \end{aligned}$$

In particular, we have $\lambda_n = \max \{ R_A(u) \mid u \in \mathbf{C}^n \setminus \{0\} \}$ and $\lambda_1 = \min \{ R_A(u) \mid u \in \mathbf{C}^n \setminus \{0\} \}$.

iv. .

Consequences : ① $\lambda_n = \max \{ R_A(u) \mid u \neq 0 \}$
 $\lambda_1 = \min \{ R_A(u) \mid u \neq 0 \}$

2. '

L15: Spectrum Graphs

Monday, October 26, 2020 10:19 PM

1. '

2. '

Mid:

Monday, October 26, 2020 10:19 PM

1. Notation

a. Simple Graphs Definition

G no loops; there is at most one edge that is incident to any pair of vertices

b. Sum of Degrees

i. Equation

$$\sum_{x \in V(G)} \deg(x) = 2 \cdot \#E(G)$$

ii. Corollary

For a finite graph G , it has an even number of vertices with odd degree

c. Walk

i. A Path

Edge of walk are distinct

ii. Closed

$$x_0 = x_n$$

iii. Simple

Vertices are distinct (automatically a path!)

iv. Simple closed

$x_0 = x_n$, while $x_0, \dots, x_{n-1}$ are distinct (this is not simple!)

2. Eulerian and Hamiltonian

a. Eulerian

i. Definition

Closed path going through all edges exactly once

ii. Theorem

Finite graph G with no isolated vertices, Eulerian **iff** G is connect and every vertex have even degree

iii. Theorem: Equivalence of Directed Eulerian Circuit

Strongly connected and in-degree = out-degree

b. Hamiltonian

i. Remark

A Hamiltonian circuit is the same as giving a spanning polygon for G

ii. Ore Theorem: A Sufficient Condition

Theorem 3 (Ore). If G is a simple graph with $n \geq 3$ vertices such that for any two non-adjacent vertices x and y , we have

$$\deg(x) + \deg(y) \geq n,$$

then G is Hamiltonian.

3. Tree

a. Three Equivalency

G is a tree (Connected graph without subgraph isomorphic to a polygon P_n)



$\forall_{x \neq y \in V(G)} \exists \text{ unique path from } x, y$



G is connected and contains no closed path (of positive length)

b. Tree and Forest

i. Tree

If G is a finitely connected graph, then $\#E(G) \geq \#V(G) - 1$, and **equality holds iff G is a tree**

ii. Forest

If G is a finite graph, then $\#E(G) \geq \#V(G) - \#\{\text{connected components of } G\}$, and **equality holds iff G is a forest**

c. Cayley's Formula

i. Multinomial

$$\binom{n}{m_1, \dots, m_k} = \frac{n!}{m_1! \dots m_k!} = \sum_{i=1}^k \binom{n-1}{m_1, \dots, m_i-1, \dots, m_k}$$

ii. 1st Proof

$$t(n; d_1, \dots, d_n) = \binom{n-2}{d_1-1, \dots, d_n-1}$$

iii. 2nd Proof

Pruffer's construction (PDF P72)

4. Depth-First Search Tree

a. Proposition 1

If x, y adjacent in G , then x is either ancestor or descendent of y in T

b. Proposition 2

i. Definition of Isthmus / Bridge

A bridge is an edge, whose deletion results in a disconnected graph

ii. Statement

For a depth-first search tree T of G with root v_0 , if x is the parent of y in T , and $\{x, y\}$ not an isthmus in G , then there is an edge $e \in E(G), e \notin E(T)$ joining a descendent a of y with ancestor b of x

5. Color

a. Brooks' Theorem

$$\left(\begin{array}{l} d \geq 3, G \text{ a finite graph without loop} \\ K_{d+1} \text{ is not isomorphic with a subgraph of } G \\ \text{All vertices of } G \text{ has degree } \leq d \end{array} \right) \Rightarrow \chi(G) \leq d$$

b. Kneser's Conjecture

i. Kneser's Graph Definition

$KG_{n,k}$ is a graph of vertex set $\binom{[n]}{k}$, where A and B are joined by an edge iff $A \cap B = \emptyset$

ii. Kneser's Bound for $\chi(KG_{n,k})$

1) Statement

$$\chi(KG_{n,k}) \leq n - 2k + 2$$

2) One Such Coloring

$$c(A) = \min\{\min(x|x \in A), n - 2k + 2\}$$

6. Ramsey Theory

a. Ramsey's Theorem $N(q_1, \dots, q_s; r)$

Given $r \geq 1$ and $q_1, \dots, q_s \geq r$ there is a number N such that the following holds: given a set S with $n \geq N$ elements such that all $\binom{n}{r}$ r -subsets of S are divided into s mutually exclusive families $T_1, \dots, T_s$ ("colors"), there is i with $1 \leq i \leq s$ and some q_i -subset of S for which every r -subset is in T_i

b. Estimating Ramsey's Number

$$\begin{cases} N(p, q; 1) = p + q - 1 \\ N(p, r; r) = p \\ N(p, q; r) \leq N(N(p-1, q; r), N(p, q-1; r); r-1) + 1 \\ N(p, q; 2) \leq \binom{p+q-2}{p-1} \end{cases}$$

7. Hall and SDR

a. Hall's Marriage Theorem

A complete matching Λ from X to Y exists **iff** for any $A \subseteq X$, $\# \Gamma(A) \geq \# A$

b. SDR

Setup. Suppose that S is a finite set and $A_0, \dots, A_{n-1}$ are subsets of S . A **system of distinct representatives** (or **SDR**, for short) is given by n distinct elements $a_0, \dots, a_{n-1} \in S$ such that $a_i \in A_i$ for $0 \leq i \leq n-1$.

c. Property (H)

For sets $A_0, \dots, A_{n-1}$, $\forall k$ the union of k sets has $\geq k$ elements

d. SDS Theorem

$A_0, \dots, A_{n-1}$ satisfies (H) **iff** they have SDR

8. Planar Graph

a. Euler's Formula

For every planar graph G , $\#F - \#E + \#V = 2$

b. Corollary 1:

If G is a connected planar simple graph, with $n \geq 3$ vertices, then $\#E(G) \leq 3n - 6$

c. Corollary 2:

Let G be a connected planar simple graph. If $g \geq 3$ is such that $\#E(G) \geq g$ and G has no P_n with $n < g$, then $\#E(G) \leq \frac{g}{g-2} (\#E(G) - 2)$

d. Theorem 2

A graph G is not planar iff it contains a subgraph which is a subdivision of either K_5 or $K_{3,3}$

L16: Inclusion-Exclusion

Monday, November 2, 2020

6:27 PM

1. '

a. Inclusion-Exclusion

i. Convention

Let's talk about finite sets

ii. Notation

For finite set A , $|A|$ equals number of element of A

iii. Introduction

$$|A \cup B| = |A| + |B| - |A \cap B|$$

iv. Theorem

$$\left\{ \begin{array}{l} E_1, \dots, E_r \text{ finite set} \\ N_J := \left| \bigcap_{i \in J} E_i \right| \quad \forall J \subseteq \{1, \dots, r\} \\ N_j := \sum_{|J|=j} N_J \end{array} \right.$$

$\Downarrow$

$$|E_1 \cup \dots \cup E_r| = \sum_{1 \leq j \leq r} (-1)^{j-1} N_j$$

v. Proof

Check how many time any element is counted on both LHS and RHS

b. Application: Derangement

i. Definition

Derangement of $\{x_1, \dots, x_n\} = A$ is permutation σ of A such that $\sigma(x_i) \neq x_i \forall i$

ii. Derivation

1) Step 1: Number of Derangement

$$\{\sigma = \text{perm of } A\} \setminus \bigcup_{i=1}^n E_i = n! - |E_1 \cup \dots \cup E_n|$$

2) Step 2

$$|E_1 \cup \dots \cup E_n| =$$

c. Application: Surjective Map

i. Note

For $X = \{x_1, \dots, x_n\}$ and $Y = \{y_1, \dots, y_k\}$ we have k^n maps $X \rightarrow Y$

2. Mobius Inversion in Arbitrary Poset

a. Poset: Partially Ordered Set

i. Relation $\leq$

$$1) \quad a \leq a \quad \text{for all } a \in P$$

$$2) \quad \text{if } a \leq b \text{ and } b \leq a, \text{ then } a = b$$

$$3) \quad \text{if } a \leq b \text{ and } b \leq c, \text{ then } a \leq c.$$

ii. Fix a Poset $(P, \leq)$

We assume that $[a, b] := \{x | a \leq x \leq b\}$ is finite

iii. Example

$$1) \quad N = \mathbb{Z}_{>0} \quad \text{with } a \leq b \text{ if } \frac{b}{a} \in \mathbb{Z}$$

(order given by divisibility)

$$2) \quad P(X) = \{A | A \subseteq X\} \quad X \text{ fixed set}$$

with $A \leq B$ if $A \subseteq B$ (order given by inclusion)

iv. Incidence Algebra of P

$$A(P) = \{f: P \times P \rightarrow \mathbb{C} | f(x, y) = 0 \text{ unless } x \leq y\}$$

b. Convolution

i. Definition

$$(f * g)(x, y) := \sum_{x \leq z \leq y} f(x, z)g(z, y)$$

ii. Example: Matrix

Note: if $P = \{1, \dots, n\}$ with usual order

a function $P \times P \rightarrow \mathbb{C} \leftrightarrow$ matrix in $M_n(\mathbb{C})$

$$A(P) = \{ \text{upper-triangular matrices} \}$$

* corresp. to matrix multiplication.

$$\left(\begin{array}{l} f: P \times P \rightarrow \mathbb{C} \leftrightarrow (f(i, j))_{i, j} \\ f \in A(P) \Leftrightarrow f(i, j) = 0 \text{ if } i > j \end{array} \right)$$

iii. Property

1) Association

$$(f * g) * h = f * (g * h)$$

2) Existence of Unit

$$1(x, y) := \begin{cases} 1 & x = y \\ 0 & x \neq y \end{cases}$$

iv. Important Element

1) Special Element that has Inverse

$$\zeta := \begin{cases} 1 & x \leq y \\ 0 & x \neq y \end{cases}$$

2) Lemma: Existence and Uniqueness of Mobius Function μ

$$\mu * \zeta = 1 = \zeta * \mu$$

3) Uniqueness

$$\mu' = 1 * \mu' = (\mu * \zeta) * \mu' = \mu$$

4) Existence

在此处键入公式。

3. '

4. ''

L17: Mobius Inversion

Tuesday, November 3, 2020 10:03 AM

1. Mobius Inversion

a. Previously: Unique Property of Mobius Function

$$\sum_{x \leq z \leq y} \mu(z, y) = \sum_{x \leq z \leq y} \mu(x, z) = \begin{cases} 1 & x = y \\ 0 & x \neq y \end{cases}$$

b. Theorem: Mobius Inversion Formula

i. Theorem

$$\begin{aligned} & \forall_{b \in P} \{a \in P \mid a \leq b\} \text{ is finite, } f, g: P \rightarrow \mathbb{C} \\ & \Downarrow \\ & \left(\forall_{y \in P} g(y) = \sum_{x \leq y} f(x) \right) \Leftrightarrow \left(\forall_{y \in P} f(y) = \sum_{x \leq y} \mu(x, y) g(x) \right) \end{aligned}$$

ii. Proof

Plug in previous property

iii. Remark: Similar Theorem

$$\begin{aligned} & \forall_{a \in P} \{b \in P \mid a \leq b\} \text{ is finite, } f, g: P \rightarrow \mathbb{C} \\ & \Downarrow \\ & \left(\forall_{x \in P} g(x) = \sum_{x \leq y} f(y) \right) \Leftrightarrow \left(\forall_{x \in P} f(x) = \sum_{x \leq y} \mu(x, y) g(y) \right) \end{aligned}$$

c. Example A: Subset and Inclusion

i. Mobius Inversion Formula

$$\forall_{A \subseteq B \in \mathcal{P}(X)} \mu(A, B) := (-1)^{|B \setminus A|}$$

ii. Proof:

$$\sum_{0 \leq p \leq n} (-1)^p \binom{n}{p} \stackrel{\text{Binomial}}{=} \begin{cases} 1 & n = 0 \\ 0 & n \geq 1 \end{cases} \Rightarrow \sum_{A \subseteq C \subseteq B} (-1)^{|C \setminus A|} = \begin{cases} 1 & B = A \\ 0 & B \neq A \end{cases}$$

iii. Application: The Inclusion-Exclusion Formula on this Poset

1) Step 1: Setup

$$E_1, \dots, E_r \subseteq E, \quad X = \{1, 2, \dots, r\}, \quad P = \mathcal{P}(X) \text{ is all subsets of } X$$

2) Step 2: Corresponding $f, g: P \rightarrow \mathbb{Z}$

$$\begin{cases} f(J) := \left| \left(\bigcap_{i \in J} E_i \right) \setminus \left(\bigcup_{i \notin J} E_i \right) \right| \\ g(J) := \left| \bigcap_{i \in J} E_i \right| \end{cases}$$

3) Step 3: Lemma From Set Theory: Disjoint Union $\sqcup$

$$\bigcap_{i \in J} E_i = \bigsqcup_{K \supseteq J} \left(\left(\bigcap_{i \in K} E_i \right) \setminus \left(\bigcup_{i \notin K} E_i \right) \right)$$

4) Step 4: Apply Step 3 to "Similar Theorem"

$$\forall_{J \in P} g(J) = \sum_{K \supseteq J} f(K)$$

5) Step 5: Mobius Inversion

$$f(J) = \sum_{K \supseteq J} \mu(J, K) g(K) = \sum_{K \supseteq J} (-1)^{|K \setminus J|} g(K)$$

6) Step 6: Plug in $J = \emptyset$ into Step 5

$$f(J) = \left| E \setminus \bigcup_{i=1}^r E_i \right| = \sum_K (-1)^{|K|} \left| \bigcap_{i \in K} E_i \right|$$

7) Step 7: Decomposition of Step 6 RHS

$$\begin{cases} \sum_K (-1)^{|K|} \left| \bigcap_{i \in K} E_i \right| = \sum_{K=\emptyset} (-1)^{|K|} \left| \bigcap_{i \in K} E_i \right| + \sum_{K \neq \emptyset} (-1)^{|K|} \left| \bigcap_{i \in K} E_i \right| \\ \sum_{K \neq \emptyset} (-1)^{|K|} \left| \bigcap_{i \in K} E_i \right| = (-1)^0 |E| \end{cases}$$

8) Step 8: Combining Step 7

$$\left| \bigcup_{i=1}^r E_i \right| = \sum_{K \neq \emptyset} (-1)^{|K|} \left| \bigcap_{i \in K} E_i \right|$$

d. Example B: $\mathbb{N}^*$ and Division

i. Basics

1) Definition

$$a/b \text{ iff } \frac{b}{a} \in \mathbb{N}^*$$

2) Mobius Inverse Formula

$$\mu(n) = \begin{cases} 1 & n = 1 \\ 0 & n \text{ is not squarefree} \\ (-1)^r & n = p_1 \dots p_r \end{cases}$$

3) Claim

$$\forall_{a, b \in \mathbb{N}^*, a/b} \mu(a, b) := \mu\left(\frac{b}{a}\right) \text{ satisfy the property Unique Property of Mobius Function}$$

4) Convention

From now on, in MATH565, $\mathbb{N} := \mathbb{N}^*$

ii. Application over Euler Formula

1) Revisit Euler Formula φ

$$\varphi: \mathbb{N} \rightarrow \mathbb{N}, \quad \varphi(n) := |\{d | 1 \leq d \leq n, \gcd(d, n) = 1\}|$$

2) Evaluation of φ

a) Step 1: Definition of Set

$$E_i := \{d | 1 \leq d \leq n, P_i/d\}$$

b) Step 2: To Euler Formula

$$\varphi(n) = n - |E_1 \cup \dots \cup E_r|$$

c) Step 3: Evaluate N_J

$$N_J = \frac{n}{\prod_{i \in J} P_i}$$

d) Step 4: Inclusive-Exclusive Formula

$$\varphi(n) = n \left(1 - \frac{1}{p_1}\right) \dots \left(1 - \frac{1}{p_r}\right)$$

e) Step 5: Algebra Over Step 4

$$n \left(1 - \frac{1}{p_1}\right) \cdots \left(1 - \frac{1}{p_r}\right) = \sum_{d|n} d\mu\left(\frac{n}{d}\right)$$

f) Step 6: Conclusion

$$\sum_{d|n} \varphi(d) = n$$

L18: Geometric Lattices

Thursday, November 5, 2020 9:51 AM

1. '

a. Lattices

i. Previously: Lattices Definition

1) Lattices

A poset L with join and meet

2) Join, Smallest upper bound, $x \vee y$

$$\begin{cases} x \leq x \vee y \\ y \leq x \vee y \\ \forall_{x \leq z, y \leq z} x \vee y \leq z \end{cases}$$

3) Meet, Largest lower bound, $x \wedge y$

$$\begin{cases} x \vee y \leq x \\ x \vee y \leq y \\ \forall_{z \leq x, z \leq y} z \leq x \vee y \end{cases}$$

ii. Chain, Cover, Atom

1) Chain

a) Convention

We always assume no infinite chain in L

b) Definition

A subset of L is a chain if any two elements are comparable

c) Example

When $\leq$ is defined as divisibility, 3 and 5 are not comparable

2) Lemma: Existence of 0_L and 1_L

If L is a lattice with no infinite chains, then L contains smallest element 0_L and largest element 1_L

3) Cover

a) Definition

In any poset P , b covers a iff $a, b \in P$, $[a, b] = \{a, b\}$

b) Description

b is the element next to a

c) Atom of Lattice

$x \in L$ such that x covers 0_L

iii. Geometric Lattices

1) Definition

A lattices $(L, \leq)$ which is atomic and semidodular

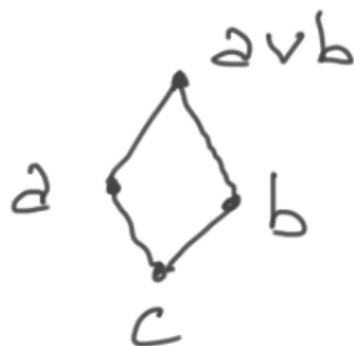
2) Atomic

Every $x \in L$ is join of finitely many atoms

3) Semi-modular

If a, b covers c , then $a \vee b$ covers both a and b

4) Scheme



5) Note

Although semi-modular doesn't always hold, $a \wedge b = c$ always hold

6) Examples

$$L = (P(X), \subseteq)$$

$$L = (\{\text{linear subspace of } V\}, \subseteq)$$

iv. Combinatorial Geometry

1) Definition

a) Notation

$$(X, F), \quad F \subseteq \{\text{subset of } X\}$$

b) Property 1: Poset Flat

F include no infinites chains (thus a poset with $\subseteq$)

c) Property 2:

$$\forall_{A, B \in F} A \cap B \in F$$

d) Property 3: $0_L, 1_L$ and Atom

$$\emptyset, X, \{x\} \in F, \quad \forall x \in X$$

e) Property 4: Partition

$\{B \setminus A \mid B \in F, B \text{ covers } A\}$ gives a partition of $X \setminus A$

f) a

b. '

L19: Combinatorial Geometries

Tuesday, November 10, 2020 10:03 AM

1. "

a. Previously: Definition

i. Geometric Lattices: A lattices $(L, \leq)$ which is atomic and semidodular

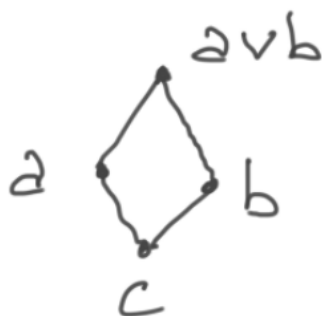
1) Atomic

Every $x \in L$ is join of finitely many atoms

2) Semi-modular

If a, b covers c , then $a \vee b$ covers both a and b

3) Scheme



4) Note

Although semi-modular doesn't always hold, $a \wedge b = c$ always hold

5) Examples

$$L = (P(X), \subseteq)$$

$$L = (\{\text{linear subspace of } V\}, \subseteq)$$

ii. Combinatorial Geometry

1) Notation

$$(X, F), \quad F \subseteq \{\text{subset}(X)\}$$

2) Property 1: Poset Flat

F include no infinites chains (thus a poset with $\subseteq$)

3) Property 2:

$$\forall_{A, B \in F} A \cap B \in F$$

4) Property 3: $0_L, 1_L$ and Atom

$$\emptyset, X, \{x\} \in F, \quad \forall x \in X$$

5) Property 4: Partition

$\{B \setminus A \mid B \in F, B \text{ covers } A\}$ gives a partition of $X \setminus A$

b. From Geometric Lattice $(L, \leq)$ to Combinatorial Geometry (X, F)

i. Basics

1) Construction

$$X = \{\text{atom in } L\}, \quad F = \{L_x \mid x \in L\}, \quad L_x = \{y \text{ atom in } L, y \leq x\}$$

2) Proposition 1:

(X, F) is a combinatorial geometry

ii. Lemma 1:

1) Statement

$$L_{x_1} \subseteq L_{x_2} \Leftrightarrow x_1 \leq x_2$$

2) Proof

To prove $\Rightarrow$, we need to apply definition of meet

iii. Proof of Proposition 1

1) Check Property 1:

Since L do not have infinite chain of $\leq$, F do not have infinite chain of $\subseteq$

2) Check Property 2:

$$L_{x_1} \cap L_{x_2} = L_{x_1 \wedge x_2} \in F$$

3) Check Property 3:

$$X = L_{1_L} \quad X \in F$$

$$\emptyset = L_{0_L} \Rightarrow \emptyset \in F$$

$$\forall y \in X \quad \{y\} = L_y \in F$$

4) Check Property 4:

We need to show that $\forall_{A \in F, x \in X \setminus A} \exists_{B \in F} B$ covers A while $x \in B$

iv. Lemma 2:

1) Statement

If $(L, \leq)$ is a geometric lattice, $x \in L$ is an atom, then for every $z \in L$ such that $\neg x \leq z$, we have $x \vee z$ covers z

c. Isomorphism

i. Definition

1) Between Poset

A bijective map $\varphi: (L_1, \leq) \rightarrow (L_2, \leq)$ such that $\forall a, b \in L, a \leq b$ iff $\varphi(a) \leq \varphi(b)$

2) Between Combinatorial Geometry $(X_1, F_1), (X_2, F_2)$

$\varphi: X_1 \rightarrow X_2$ such that for every $A \subseteq X_1$, we have $A \in F_1$ iff $\varphi(A) \in F_2$

ii. Conclusion

The posets and combinatorial geometries are equivalent

d. Example: $AG_n(\mathbb{F})$

i. Affine Linear Subspace

1) Definition A

$$A = W + a := \{w + a | w \in W\}$$

2) Remark

$$W \text{ is determined by } A. W = \{a_1 - a_2 | a_1, a_2 \in A\}$$

ii. Definition: $AG_n(\mathbb{F})$

$(V, \{\emptyset\} \cup \{\text{affine linear subspace of } V\})$ is a combinatorial geometry.

e. Properties of Combinatorial Geometries

i. Definition: $\bar{A}$

$$\bar{A} := \bigcap_{B \in F, A \subseteq B} B$$

ii. Remark

$$\bar{A} \in F, \quad A \subseteq \bar{A}$$

iii. Properties

$$1) \text{ If } A_1 \subseteq A_2 \subseteq X \Rightarrow \bar{A}_1 \subseteq \bar{A}_2$$

$$2) \text{ If } A \in F \Rightarrow \bar{A} = A.$$

In particular: for every $B \in F, \bar{\bar{B}} = \bar{B}$

$$3) \text{ If } A, B \in F \Rightarrow A \vee B = \overline{A \cup B}$$

↑
lattice-theoretic join.

iv. Example

$A \subseteq \mathbb{F}^n$, $\bar{A}$ is an affine linear span of A

v. Independent

A subset $S \subseteq X$ is independent if $\forall x \in S, x \notin \overline{S \setminus \{x\}}$ (in other words, $\forall x \in S$ there is no flat $A \in F$ such that $S \setminus \{x\} \subseteq A$ but $x \notin A$)

vi. Lemma 3: Exchange Axiom

For $A \subseteq X$ and $x, y \in X$, if $x \notin \bar{A}$ but $x \in \overline{A \cup \{y\}}$, then $y \in \bar{A \cup \{x\}}$

2. '

1. '

a. Previously

i. Definition

1) Smallest Flat Containing A

$$\bar{A} := \bigcap_{B \in F, A \subseteq B} B$$

2) Independency

A subset $S \subseteq X$ is independent if $\forall x \in S, x \notin \overline{S \setminus \{x\}}$ (in other words, $\forall x \in S$ there is no flat $A \in F$ such that $S \setminus \{x\} \subseteq A$ but $x \notin A$)

ii. Exchange Axiom

For $A \subseteq X$ and $x, y \in X$, if $x \notin \bar{A}$ but $x \in \overline{A \cup \{y\}}$, then $y \in \overline{A \cup \{x\}}$

b. Correlations

i. Correlation 1

1) Statement

If $S \subseteq X$ is independent and $x \in X \setminus \bar{S}$, then $S \cup \{x\}$ is independent

2) Proof

a) Step 1:

We only need to show $\forall y \in S, y \notin \overline{(S \setminus \{y\}) \cup \{x\}}$

b) Step 2:

If step 1 does not hold, then according to exchange axiom, $x \in \overline{(S \setminus \{y\}) \cup \{y\}} = \bar{S}$, which is a contradiction

3) Remark:

If $S \subset T$ are two independent set, then $\bar{S} \subset \bar{T}$

4) Proof

a) Step 1: Properties of $\bar{A}$

$\bar{S} \subseteq \bar{T}$. We need to show $\bar{S} \neq \bar{T}$

b) Step 2: Distinct

$x \in T \setminus S$, then $x \in \bar{T}$. Since $S \subseteq T \setminus \{x\}$ and $x \notin \overline{T \setminus \{x\}}$, thus $x \notin \bar{S}$

5) Further Remark: No Infinite Sequence of Independent Set

Because there is no infinite strictly increasing sequence of flat

ii. Correlation 2

1) Statement

$S \subseteq A \subseteq X$ and S is a maximal independent subset of A , then $\bar{S} = \bar{A}$

2) Proof

a) Step 1: $\bar{A}$ properties

We have $\bar{S} \subseteq \bar{A}$. We need to show that $\emptyset = A \setminus \bar{S}$

b) Step 2: Contradiction of Maximality

Suppose $x \in A \setminus \bar{S}$, then by cor.1, $S \cup \{x\}$ is independent. Thus S is not maximal

3) Proposition 1

For $A \subseteq X$, any two maximal independent subset of A have the same number of element

4) Proof

I don't really understand

c. Basis and Rank

i. Definition

1) A Generating Set of A

The subset $S \subseteq A$ such that $\bar{S} = A$

2) A Basis of A

Maximal independent subset $S \subseteq A$

3) Rank of A : $rk(A)$

$$rk(X, F) = rk(X)$$

4) Terminology (Motivated by Affine Geometry)

	rk
Points	1
Lines	2
Planes	3
Hyperplanes	$rk(X, F) - 1$

ii. Remark

1) Remark 2: Basis and Generating Set

S is a basis of A iff S is an independent generating set

2) Remark 3: Basis and Independent Subset

For every $A \subseteq X$, if $S \subseteq A$ is a maximal independent subset of A , then S is a basis for $\bar{A}$

3) Remark 4: Subset and Generating Set

$A \in F$, a subset $S \subseteq A$ is a basis iff it is a minimal generating set

4) Remark 5: Rank and Span

If $A \subseteq B$ are two flats, then $rk(A) \leq rk(B)$, with equality iff $A = B$

5) Remark 6: Cover and Rank

If $A, B \in F$ and $A \subseteq B$, then B covers A iff $rk(B) = rk(A) + 1$

6) Remark 7: Special Case for 6

$$rk(A) = 1 \Leftrightarrow A \text{ covers } \emptyset \Leftrightarrow A = \{x\}, x \in X$$

7) Remark 8: Length of Chain

$A, B \in F$, chain $A = A_0 \subset A_1 \subset \dots \subset A_r = B$ such that A_i covers A_{i-1} , then $r = rk(B) - rk(A)$

iii. Proposition 2

1) Statement: Semi-modular Law

$$\forall_{A, B \in F} rk(A \wedge B) + rk(A \vee B) \leq rk(A) + rk(B)$$

2) Proof

a) Step 1: Lattice Notation in $AG_n(\mathbb{F})$

$$A \wedge B = A \cap B, \quad A \vee B = \overline{A \cup B}$$

b) Step 2:

Choose S a basis of $A \cap B$, and complete S to basis S_1 of A and S_2 of B

c) Step 3: Cardinalities

$$r := |S|, \quad r + n_i := |S_i|, \quad i = 1, 2$$

d) Step 4: Span

$$A \vee B = \overline{A \cup B} = \overline{S_1 \cup S_2} \Rightarrow S_1 \cup S_2 \text{ is a generating set of } A \vee B$$

e) Step 5: Remark 4

$$|S_1 \cup S_2| \geq rk(A \vee B)$$

f) Step 6: Inclusive-Exclusive

$$|S_1 \cup S_2| = |S_1| + |S_2| - |S_1 \cap S_2| \leq rk(A) + rk(B) - rk(A \wedge B)$$

g) Step 7: Combining Step 5 and Step 6

Q.E.D

3) Modula

(X, F) is called modular if for every flat A, B , we have equality in Semi-modular law

a. '

L21: Geometric Lattice and Graph

Thursday, November 19, 2020 9:40 AM

1. '

a. Previously

i. Special Element and Mobius Inverse

$$\zeta := \begin{cases} 1 & x \leq y \\ 0 & x \neq y \end{cases}, \quad \sum_{x \leq z \leq y} \mu(z, y) = \sum_{x \leq z \leq y} \mu(x, z) = \begin{cases} 1 & x = y \\ 0 & x \neq y \end{cases}$$

b. Mobius Functions for Lattices

i. Proposition 1

1) Statement

If $(L, \leq)$ is a finite lattice and $a \in L, a \neq 0_L$, then $\sum_{\substack{x \in L \\ a \vee x = 1_L}} \mu(0_L, x) = 0$

2) Proof

Calculate following sum in two ways

$$\sum_{x, y \in L} \mu(0_L, x) \zeta(x, y) \eta(a, y) \zeta(x, y)$$

3) Corollary

Suppose L is a finite geometric lattice, for every $x \leq y$ in L , we have $\mu(x, y) \neq 0$ and its sign is $(-1)^{rk(y) - rk(x)}$

c. Review about Equivalence Relation

i. Convention

Suppose X is a set

ii. Points

1) Relation

A relation on X is a subset $R \subseteq X \times X$, we write xRy if $(x, y) \in R$

2) Inclusion Order

Order can be given by set inclusion

3) Equivalence Relation $\sim$

$$\begin{cases} x \sim x \quad \forall x \in X \\ x \sim y \Rightarrow y \sim x \\ x \sim y, y \sim z \Rightarrow x \sim z \end{cases}$$

4) Partition: A Disjoint Union

$$X = \bigcup_{i \in I} A_i$$

5) Equivalence and Partition

$x \sim y$ iff $x, y \in$ same A_i

6) Smallest Equivalence Relation S for Any Relation R such That $R \leq S$

For any $x, y \in X$, we put xSy if there is a sequence $x = x_0, x_1, \dots, x_n = y$ with $n \geq 0$ such that for $i \geq 1$, we have either $x_{i-1}Rx_i$ or x_iRx_{i-1}

7) Compare Equivalence

$$R: \quad X = \bigsqcup_{i \in I} A_i, \quad S: \quad X = \bigsqcup_{j \in J} B_j$$

$$\begin{aligned} R \leq S &\iff \forall i, A_i \text{ is contained in some } B_j \\ &\iff \forall j, B_j \text{ is a union of } A_i\text{'s.} \end{aligned}$$

2. Geometric Lattice $L(G)$ Corresponding to a Graph G

a. Basics

i. Definition

For finite simple graph $G, X = V(G), L(G) \subseteq \{\text{partitions of } X\} = \{A \mid \text{the subgraph of } G \text{ spanned by each block of } A \text{ is connected}\}$

ii. Order Relation

$A \leq B$ iff every block of A is contained in a block of B

iii. Smallest Element and Largest Element

$0_{L(G)}$	Partition in singletons
$1_{L(G)}$	Partition with the connected components of G

b. Color Function with Fixed $x \in \mathbb{N}^*$

i. Definition

$g(A)$	numbers of colorings of X with x colors such that all vertices in the same block have the same colors	$x^{ A }$
$f(A)$	Numbers of colorings of X with x colors such that all vertices in the same vlock get same color, and any two vertices in different blocks which are neighbours in G gets different color	

ii. Proof

1) Step 1: Observation

Given A and a coloring "counted by $g(A)$ ", then there is a unique $B \geq A$ such that the coloring is "counted by $f(B)$ ". Such B is obtained from A by putting together blocks of A_i and A_j in A colored by the same color and such that we have some $x_i \in A_i, x_j \in A_j$ adjacent in G

2) Step 2: Formulize Step 1

$$g(A) = \sum_{A \leq B} f(B)$$

3) Step 3: Apply Mobius Inversion

$$f(A) = \sum_{A \leq B} \mu_{L(G)}(A, B) g(B) = \sum_{A \leq B} \mu_{L(G)}(A, B) x^{|B|}$$

4) Step 4: Proper Coloring, $f(0_{L(G)})$

$f(0_{L(G)})$ is the number of proper coloring

iii. Conclusion: Chromatic Polynomial

$$Q(x) = \sum_{B \in L(G)} \mu_{L(G)}(0_{L(G)}, B) x^{|B|}$$

c. Proposition 2

i. Statement

For every finite simple graph G , $L(G)$ is a geometric lattice

ii. Proof of Lattice

1) Definition

$$A, B \in L(G), \quad X = \bigcup_{i \in I} A_i = \bigcup_{j \in J} B_j$$

2) Meet

$A \wedge B$: consider for every $A_i \cap B_j$ nonempty the connected component of the subgraph of G spanned by the vertices in $A_i \cap B_j$

3) Join

$A \vee B$: consider the partition defined as follows
if A corresp to equiv rel R_A
 B — " — R_B

take $A \vee B$ to corresp to the smallest equiv rel that's $\geq R_A \cup R_B \subseteq X \times X$.

iii. Proof of Geometric Lattice

1) No infinite Chain

Clearly

2) Atoms $R_{\{x,y\}}$

Partition where only one block is given by $\{x, y\}$ where x, y neighbors. Note that atoms are in bijection with $E(G)$

3) Atomic

Given $A \in L(G)$, A is the join of atoms $R_{\{x,y\}}$ where x, y are in the same block of A and are neighbours in G

4) Covers

B covers A iff B is obtained from A by putting together two blocks A_i and A_j of A such that A_i and A_j have adjacent vertices

5) Semi-modular

Case 1 B is obtained by putting together A_i, A_j
 $\mathcal{C}$ — " ————— A_i, A_k $j \neq k$

Case 2 B — " ————— A_i, A_j
 $\mathcal{C}$ — " ————— A_k, A_e

In case 1: $B \vee \mathcal{C}$ is obtained ^{from A} by putting together A_i, A_j, A_k .

In case 2: $B \vee \mathcal{C}$ is obtained from A by putting together A_i, A_j, A_k, A_e .

Exer: check that $B \vee \mathcal{C}$ covers both B and $\mathcal{C}$.

6)

3. '

L22: Lattice of Vector Subspace Over Finite Field

Monday, November 30, 2020 9:16 PM

1. Geometric Lattice $L(G)$ Corresponding to Graph G

a. Rank

Since $0_{L(G)}$ has $n = |V(G)|$ blocks, thus for every $A \in L(G)$, $\text{rank } A = n - |A|$

b. Numbers of Proper Coloring $Q(x) \in \mathbb{Z}[x]$

$$Q(x) = \sum_{B \in L(G)} \mu_{L(G)}(0_{L(G)}, B) x^{n - \text{rank } B}$$

c. Special Case: Connected G

$$\forall B \in L(G), \quad 0 \leq \text{rank } B \leq n - 1$$

$$\text{Thus } Q(x) = x^n - a_1 x^{n-1} + \dots + (-1)^{n-1} a_{n-1} x, \quad a_i = \sum_{\text{rank}(B)=i} \mu(0, B) \times (-1)^i > 0$$

2. q -binomial Coefficients

a. Basics

i. Definition: Rational Function about Variable q

$$\begin{bmatrix} m \\ k \end{bmatrix}_q := \begin{cases} \frac{(q^m - 1)(q^{m-1} - 1) \dots (q - 1)}{(q^k - 1) \dots (q - 1)(q^{m-k} - 1) \dots (q - 1)} = \frac{(q^m - 1) \dots (q^{m-k+1} - 1)}{(q^k - 1) \dots (q - 1)} & k \leq m \\ 0 & k > m \end{cases}$$

ii. Remark

$$\begin{bmatrix} m \\ k \end{bmatrix}_q = \begin{bmatrix} m \\ m-k \end{bmatrix}_q, \quad \begin{bmatrix} m \\ 0 \end{bmatrix}_q = 1, \quad \begin{bmatrix} m \\ 1 \end{bmatrix}_q = \frac{q^m - 1}{q - 1} = q^{m-1} + \dots + q + 1$$

b. Lemma 1: Recursion Formula

i. Formula

$$\begin{bmatrix} m \\ k \end{bmatrix}_q = \begin{bmatrix} m-1 \\ k \end{bmatrix}_q + q^{m-k} \begin{bmatrix} m-1 \\ k-1 \end{bmatrix}_q \quad \forall m, k \in \mathbb{N}^*$$

ii. Prove

Expand and rearrange RHS

c. Proposition 1

i. Statement

For all $m, k \geq 0$, we have $\begin{bmatrix} m \\ k \end{bmatrix}_q \in \mathbb{Z}[q]$

ii. Proof

Pf: We do induction on $m \geq 0$.

If $m = 0 \Rightarrow$ this is $\begin{cases} 0 & \text{if } k > 0 \\ 1 & \text{if } k = 0 \end{cases}$ ok

Suppose now $m > 0$. If $k = 0 \Rightarrow$ this is 1 ok

If $k > 1$: $\begin{bmatrix} m \\ k \end{bmatrix}_q = \begin{bmatrix} m-1 \\ k \end{bmatrix}_q + q^{m-k} \begin{bmatrix} m-1 \\ k-1 \end{bmatrix}_q \in \mathbb{Z}[q]$
by the lemma by induction $\square$

d. Examples: $q = 1$: Return to Binomial

$$k \leq m \Rightarrow \left[\begin{matrix} m \\ k \end{matrix} \right]_2 \Big|_{2=1} = \lim_{2 \rightarrow 1} \frac{\frac{2^m - 1}{2 - 1} \cdot \dots \cdot \frac{2 - 1}{2 - 1}}{\frac{2^k - 1}{2 - 1} \cdot \dots \cdot \frac{2 - 1}{2 - 1} \cdot \frac{2^{m-k} - 1}{2 - 1} \cdot \dots \cdot \frac{2 - 1}{2 - 1}}$$

$$= \frac{m \cdot (m-1) \cdot \dots \cdot 1}{k \cdot (k-1) \cdot \dots \cdot 1 \cdot (m-k) \cdot (m-k-1) \cdot \dots \cdot 1} = \frac{m!}{k! (m-k)!} = \binom{m}{k}$$

e. Lemma 2

i. Formula: For $m \geq 1$, we have

$$\prod_{k=0}^{m-1} (1 + q^k t) = \sum_{k=0}^m q^{\frac{k(k-1)}{2}} \left[\begin{matrix} m \\ k \end{matrix} \right]_q t^k$$

ii. Remark: $q = 1$

$$(1 + t)^m = \sum_{k=0}^m \binom{m}{k} t^k$$

iii. Proof

Induction over m

3. The Lattice of Vector Subspaces of A Vector Space Over a Finite Field

a. Basics

i. Finite Field F

$F = \mathbb{Z}_p$, p is a prime number

ii. Abuse of Notation

Let $q = |F|$

iii. Fact

$q = p^e$ for some prime p and some $e \geq 1$

iv. V Be a Finite Dimensional Vector Space over F

Notice that V is isomorphic to F^n

v. Remark

We saw that the lattice $L(V)$ is a geometric lattice

b. Proposition 2: Number of Subspace

i. Statement

For every $k \leq n$, the number of k - dimensional vector subspace of V is $\left[\begin{matrix} n \\ k \end{matrix} \right]_q$

ii. Proof

1) Step 1: Count Number of Chains in V

$0 \subseteq V_1 \subseteq V_2 \subseteq \dots \subseteq V_n = V$ where $\dim V_i = i$

2) Step 1.1: Choice of V_1

$$\frac{|\{u \in V | u \neq 0\}|}{\text{multiplier by element of } F \setminus \{0\}} = \frac{q^n - 1}{q - 1}$$

3) Step 1.3: Given Above V_1 , Count Choice of V_2

$$\frac{|\{u \in V \setminus V_1 | u \neq 0\}|}{\text{multiplier by element of } F \setminus \{0\}} = \frac{q^{n-1} - 1}{q - 1}$$

4) Step 1.4: Number of Chain $M(n, q)$

$$M(n, q) = \frac{(q^n - 1)(q^{n-1} - 1) \dots (q - 1)}{(q - 1)^n}$$

5) Step 2: Construct Function φ

$\varphi: \{\text{chains } 0 \subseteq V_1 \subseteq V_2 \subseteq \dots \subseteq V_n = V \text{ where } \dim V_i = i\} \rightarrow \{k - \dim \text{ subspace of } V\}$

6) Step 2.1: Surjective

φ is surjective, for every $W \in RHS$, we have

$$|\varphi^{-1}(W)| = |\{\text{chains as above in } W\}| \cdot |\{\text{chains as above in } V \setminus W\}| = M(k, q) \cdot M(n - k, q)$$

7) Step 2.2: Rearrangement

$$|\{k - \text{dim subspace of } V\}| = \frac{M(n, q)}{M(k, q) \cdot M(n - k, q)} = \begin{bmatrix} n \\ k \end{bmatrix}_q$$

iii. Proposition 3: Mobius Function

1) Statement

For V a vector space over F as above, and $U \subset W$ subspaces, we have

$$\mu_{L(V)}(U, W) = (-1)^k q^{\frac{k(k-1)}{2}}, \quad k := \dim W - \dim U$$

2) Remark

Result from last time gives information about sign

3) Proof

a) Step 1: Plug into Property of Mobius Function

$$\sum_{k=0}^m \sum_{\substack{U \subset U' \subset W \\ \dim U' - \dim U = k}} (-1)^k q^{\frac{k(k-1)}{2}} = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$

b) Step 2: Numbers of U'

U' varies over the subspace of W/U of k dimension. From Prop 2, we know there are $\begin{bmatrix} m \\ k \end{bmatrix}_q$ of them

c) Step 3: Plug in

$$LHS = \sum_{k=0}^m (-1)^k \begin{bmatrix} m \\ k \end{bmatrix}_q q^{\frac{k(k-1)}{2}} \stackrel{\text{lemma 2}}{=} \left. \prod_{j=0}^{m-1} (1 + q^j t) \right|_{t=-1}$$

d) Step 4: Evaluate Step 3

$$\left. \prod_{j=0}^{m-1} (1 + q^j t) \right|_{t=-1} = \left. \prod_{j=1}^{m-1} (1 + q^j t) \right|_{t=-1} \times (1 + q^0 t) \Big|_{t=-1} = 0$$

c. '

d. '

e. '

4. '

5. '

6. '

7. '

1. Matroid

a. Basics

i. Definition

1) Matroid M

A matroid M is a pair (E, I)

2) Ground Set E

E is a finite set

3) Independent Sets I

I is a collection of subset of E satisfying following condition

4) Independent Conditions

$$\begin{cases} \emptyset \in I \\ \text{If } A \subseteq B \text{ and } B \in I \Rightarrow A \in I \\ \text{If } A, B \in I, \text{ such that } |A| > |B|, \text{ then there is } a \in A \setminus B \text{ such that } B \cup \{a\} \text{ independent} \end{cases}$$

5) Equivalent

Two matroids are equivalent if there is a bijection between the 2 ground sets that preserves the independent subsets

ii. Example

1) Construction

V is a finite-dimension vector space over a field k . If $E \subseteq V$ is a finite subset and $I = \{A \subseteq E \mid A \text{ linearly independent}\}$

2) Representable Matroid over k

Any matroid equivalent to a matroid (E, I) as above is a representable matroid over k

b. More Definition

i. Basis

1) Definition

A basis of M is a maximal independent subset

2) Comment

Every independent set is contained in a basis. Due to exchange axiom, any two basis have same number of element. Such number is denoted as rank M

ii. Dependent Set

A subset of E is dependent if it is not in I

iii. Circuit

A circuit of M is a minimal dependent subset of E

iv. Equivalent Expression

To give a matroid, we may give (instead of I) either of the following: all basis of M ; all dependent sets; all circuits

v. Child Matroid

If $M = (E, I)$ is a matroid, and $A \subseteq E$, then $(A, \{X \mid X \subseteq A, X \in I\})$ is again a matroid with rank A

vi. Flat B

A flat of M is a subset $B \subseteq E$ such that for every $b \notin B$, $\text{rank } B < \text{rank}(B \cup \{b\})$

vii. Set of Flats $L(M)$

$L(M) := \{\text{flats of } M\}$. Our goal is to show $(L(M), \subseteq)$ is a geometric lattice

c. Development

i. Lemma 1

1) Statement

If A, B are flats in E , then $A \cap B$ is a flat

2) Remark

Since E is finite, we can find a maximal subset $B \subseteq E$ with $A \subseteq B$ and $\text{rank } B = \text{rank } A$

ii. $\bar{A}$

1) Definition

$\bar{A} := \bigcap_{\substack{A \subseteq A' \\ A' \text{ flat}}} A'$, this is a flat according to Lemma 1

2) Property 1

$\bar{A}$ is the smallest flat containing A

3) Property 2:

If $A \subseteq B$, then $\bar{A} \subseteq \bar{B}$

4) Property 3:

$\bar{A} = A$ iff A is a flat

iii. Lemma 2

1) Statement

$(L(M), \subseteq)$ is a lattice

2) Meet

$A \wedge B = A \cap B$

3) Join

$A \vee B = \overline{A \cup B}$

iv. Lemma 3

For every $A \subseteq E$, we have $\text{rank } A = \text{rank } \bar{A}$

v. Lemma 4

If $A, B \in L(M)$ are such that B covers A , then $\text{rank } B = \text{rank } A + 1$

vi. Lemma 5

For every $A \subseteq E$, we have $\bar{A} = \{a \in E \mid \text{rank}(A \cup \{a\}) = \text{rank}(A)\}$

d. ????????

i. Definition

1) Loop

For matroid $M = (E, I)$, an element $x \in E$ is a loop if $\{x\}$ is dependent

2) Parallel

Two distinct $x, y \in E$ which are not loops are parallel if $\text{rank}\{x, y\} = 1$

3) Simple Matroid

M is a simple matroid if it has no loops and no parallel pairs of points. Note that M is a simple matroid iff $\forall x, y \in E, \{x, y\}$ independent

ii. Example: Finite Dimension Vector Space over k

1) Construction

$E \subseteq V$ is a finite subset, $M = (E, \text{linear independent subsets of } E)$

2) Property 1:

$x \in E$ is a loop iff $x = 0$

3) Property 2:

Given any two nonzero distinct $x, y \in E$, they are parallel iff they are scalar multiplier of each others

iii. Simple Matroid Associated to an Arbitrary Matroid

1) Construction

Let $M = (E, I)$ a matroid

2) Equivalence Relation $\sim$

Define an equivalence relation on E by $x \sim y$ if $\overline{\{x\}} = \overline{\{y\}}$

3) Equivalent Class $[x]$

$[x] := \{y \mid y \sim x\}$

4) Set of Equivalence Class $\hat{E}$

$\hat{E} = \{[x] \mid x \notin \bar{\emptyset}\}$ (this means x is not in any loop)

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2. Simple Matroid Associated to an Arbitrary Matroid

a. Previously

i. Construction

Let $M = (E, I)$ a matroid

ii. Equivalence Relation $\sim$

Define an equivalence relation on E by $x \sim y$ if $\overline{\{x\}} = \overline{\{y\}}$

iii. Equivalent Class $[x]$

$[x] := \{y | y \sim x\}$

iv. Set of Equivalence Class $\hat{E}$

$\hat{E} = \{[x] | x \notin \bar{\emptyset}\}$ (this means x is not a loop)

v. Definition of Independence

A subset $\{[x_1], \dots, [x_r]\}$ is independent iff $\{x_1, \dots, x_r\} \in I$

vi. Result

$\hat{M} := (\hat{E}, \hat{I})$ is a simple matroid

vii. Facts

$L(M)$ isomorphsim to $L(\hat{M})$

b. Proposition 1

If $M = (E, I)$ is a simple matroid, then $(E, L(M))$ is a finite combinatorial geometry such that I is the set of independent sets in this combinatorial geometry

c. Proposition 2

If (E, F) is a finite combinatorial geometry and I is the set of independent sets, then (E, I) is a simple matroid such that the set of flats is F . Moreover, the two notions of rank coincide on the subsets of E

d. Examples: Graph Matroids

i. Construction

Let G be a finite graph, consider the set $E(G)$.

ii. Independent

$A \subseteq E(G)$ is independent if the graph G_A with vertex set $V(G)$ and edge set A is a forest. (This means $|E(G_A)| = |V(G_A)| - |\text{connected component of } G_A|$)

iii. Claim

$(E(G), \text{independent subsets})$ is a matroid $(M(G))$

iv. Remark

$a \in E(G)$ is a loop iff a is a loop in G

If $a \neq b \in E(G)$ are not loops, then a and b are parallel in $M(G)$ iff a, b joins the same points

v. Claim

G is a simple graph iff $M(G)$ is a simple matroid

vi. Claim

$L(G)$ is isomorphic to $L(M(G))$

3. The Characteristic Polynomial of a Matroid

a. Basics

i. Construction

Let M be a simple matroid with lattice of flats $L(M)$.

ii. The characteristic polynomial of M

$$\chi(M; t) = \sum_{a \in L(M)} \mu_{L(M)}(0, a) t^{\text{rank } M - \text{rank } a}$$

iii. Examples: Simple Graph with Chromatics Polynomial $Q(t)$

$$Q(t) = \sum_{A \in L(G)} \mu_{L(G)}(0, A) t^{|A|}, \quad |A| = n - \text{rank } A, \quad n = |V(G)|, \quad \text{thus } Q(t) = t \chi(M(G), t)$$

b. Mobius Algebra of A Lattice

i. Basics

1) Construction

Let $(L, < p =)$ be a finite poset having joins $\vee$

2) Indexing the Basis

Let $A(L)$ be a real vector space with basis $\{e_x | x \in L\}$, define $A(L) = \{\sum_{x \in L} \lambda_x e_x | \lambda_x \in \mathbb{R} \forall x \in L\}$

3) Dimension

$$\dim A(L) = |L|$$

4) Multiplication with Distributivity

$$e_x \cdot e_y := e_{x \vee y}, \quad \left(\sum_{x \in L} a_x e_x \right) \cdot \left(\sum_{y \in L} b_y e_y \right) := \sum_{x, y \in L} a_x b_y e_{x \vee y}$$

5) Definition of σ_x

$$\sigma_x := \sum_{x < p = y} \mu(x, y) e_y, \quad \mu := \mu_L$$

ii. Lemma 1:

1) Statement

$$e_x = \sum_{x < p = y} \sigma_y$$

2) Proof

$$\begin{aligned} \sum_{y; x \leq y} \sigma_y &= \sum_{y; x \leq y} \sum_{z; y \leq z} \mu(y, z) e_z \\ &= \sum_{z; x \leq z} \left(\sum_{y; x \leq y \leq z} \mu(y, z) \right) e_z \\ &= \begin{cases} 1 & \text{if } x = z \\ 0 & \text{if } x \neq z \end{cases} \\ &= e_x \quad \square \end{aligned}$$

3) Corollary 1

$\{e_x | x \in L\}$ is a basis for $A(L)$

4) Proposition 3

a) Statement

$$\forall_{x, y \in L} \sigma_x \cdot \sigma_y = \delta_{x, y} \sigma_x, \quad \delta_{x, y} = \begin{cases} 1 & x = y \\ 0 & x \neq y \end{cases}$$

b) Isomorphism Vector Space

Let $\mathbb{R}^L$ be the real vector space of all tuples of real numbers $(a_x)_{x \in L}$ indexed by $x \in L, a_x \in \mathbb{R}$. The multiplication was defined component-wise

c) Standard Basis $(\alpha_x)_{x \in L}$

$$(\alpha_x)_y = \begin{cases} 1 & x = y \\ 0 & x \neq y \end{cases}, \quad \text{note } \alpha_{x_1} \cdot \alpha_{x_2} = \delta_{x_1, x_2} \alpha_{x_1}$$

d) Map $\varphi: A(L) \rightarrow \mathbb{R}^L$

$$\varphi \left(\sum_{x \in L} a_x e_x \right) = \sum_{x \in L} a_x \varphi(e_x) \text{ where } \varphi(e_x) = \sum_{x < p = u} \alpha_u$$

e) Property of φ

$$\text{A linear transformation with multiplication } \varphi(e_x \cdot e_y) = \varphi(e_x) \varphi(e_y)$$

L25: Cross-Cut Theorem

Saturday, December 12, 2020 5:32 PM

1. "

a. '

i. Formula

$$\forall_{x \in L} \varphi(\sigma_x) = \alpha_x$$

ii. Conclusion

φ maps a basis of $A(L)$ to a basis of $x \in L$, thus φ is a linear isomorphism

iii. Proposition

Since φ is invertible and preserve products, $\sigma_x \cdot \sigma_y = \delta_{x,y} \sigma_x$

b. Cross-Cut Theorem

i. Theorem 1:

1) Statement

Let L be a finite lattice with 0_L and 1_L . Suppose $X \subseteq L$ such that $0_L \notin X$ while all atoms in L are in X , then $\mu(0_L, 1_L) = \sum_{i \geq 0} (-1)^i N_i$, $N_i := \# \{I \subseteq X \mid \#I = i, \forall_{j \in I} j = 1_L\}$

2) Proof

$$\prod_{x \in X} (e_{0_L} - e_x) = \prod_{x \in X} \left(\sum_{y: \neg x \leq y} \sigma_y \right)$$

ii. Application: Characteristic Polynomial of A Simple Matroid

1) Revisit Simple Matroid Case

$$\chi(M; t) = \sum_{A \in L(M)} \mu(\emptyset, A) t^{\text{rank } M - \text{rank } A}$$

2) Proof

a) Step 1: Consider the Closed Interval

$$[\emptyset, A] \subseteq L(M), \quad X := \{\{x\} \mid x \in A\}$$

b) Step 2: Cross-Cut Theorem

$$\mu(\emptyset, A) = \sum_{\substack{B \subseteq A \\ A=B}} (-1)^{\#B}$$

c) Step 3: Combining Step 2 with Revisit

$$\chi(M; t) = \sum_{A \in L(M)} \left(\sum_{\substack{B \subseteq A \\ A=B}} (-1)^{\#B} \right) t^{\text{rank } M - \text{rank } A}$$

3) Result

$$\chi(M; t) = \sum_{B \subseteq E} (-1)^{\#B} t^{\text{rank } M - \text{rank } A}$$

4) Comment

We thus define the non-simple case polynomial the same

c. Deletion-Contraction

i. Deletion \

1) Restriction: $A \subseteq E$

$$M|_A = (A, \{B \mid B \in I, B \subseteq A\})$$

2) Deletion: If $e \in E$

$$M \setminus e = M|_{E \setminus \{e\}}$$

3) Graph Example

$$M(G) \setminus e = M(G \setminus e)$$

ii. Contraction /

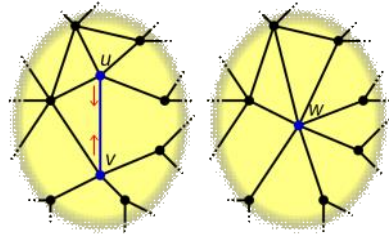
1) Assumption

$e \in E$ is not a loop

2) Definition

E/e is a matroid with ground set $E \setminus \{e\}$ and $A \subseteq E \setminus \{e\}$ is independent iff $A \cup \{e\} \in I$

3) Graph Example



iii. Coloop

1) Definition

$e \in E$ is a coloop in M iff e lies in every basis of E

2) Comment

Coloop is named so because it is a loop in the dual matroid

iv. Deletion-Contraction Formula

1) Statement: $M = (E, I)$ a Matroid and $e \in E$

$$\begin{cases} \chi(M, t) = \chi(M \setminus e, t) - \chi(M/e, t) & e \text{ not a loop or coloop} \\ \chi(M, t) = 0 & e \text{ a loop} \\ \chi(M, t) = (t - 1)\chi(M \setminus e, t) & e \text{ a coloop} \end{cases}$$

2) Comment

The Formula gives a procedure to compute $\chi(M, t)$

3) Proof: Easy Property

$$\text{rank}_{M \setminus e} A = \text{rank}_M A$$

4) Proof: If e is a Coloop

$$\text{rank } M \setminus e = \text{rank } M - 1$$

2. '

1. '
2. Basics

a. Inclusion-Exclusion Formula

i. Formula

$$|E_1 \cup \cdots \cup E_r| = \sum_{1 \leq j \leq r} (-1)^{j-1} N_j, \quad \begin{cases} E_1, \dots, E_r \text{ finite set} \\ N_j := \left| \bigcap_{i \in J} E_i \right| \quad \forall J \subseteq \{1, \dots, r\} \\ N_j := \sum_{|I|=j} N_I \end{cases}$$

ii. Examples

1) Derangement

$$\{\sigma = \text{perm of } A\} \setminus \bigcup_{i=1}^n E_i = n! - |E_1 \cup \cdots \cup E_n|$$

2) Surjective Map

For $X = \{x_1, \dots, x_n\}$ and $Y = \{y_1, \dots, y_k\}$ we have k^n maps $X \rightarrow Y$
- b. Mobius Function

i. Statements

1) Unit and Special Element

$$1(x, y) := \begin{cases} 1 & x = y \\ 0 & x \neq y \end{cases}, \quad \zeta := \begin{cases} 1 & x \leq y \\ 0 & x \not\leq y \end{cases}$$

2) Mobius Inversion

$$\sum_{x \leq z \leq y} \mu(z, y) = \sum_{x \leq z \leq y} \mu(x, z) = \begin{cases} 1 & x = y \\ 0 & x \neq y \end{cases}$$

3) Equivalent Condition

$$\begin{matrix} \forall a \in P \{a \leq b\} \text{ is finite, } f, g: P \rightarrow \mathbb{C} \\ \Downarrow \\ \left(\forall_{y \in P} g(y) = \sum_{x \leq y} f(x) \right) \Leftrightarrow \left(\forall_{y \in P} f(y) = \sum_{x \leq y} \mu(x, y) g(x) \right) \end{matrix} \quad \begin{matrix} \forall a \in P \{b \in P | a \leq b\} \text{ is finite, } f, g: P \rightarrow \mathbb{C} \\ \Downarrow \\ \left(\forall_{x \in P} g(x) = \sum_{x \leq y} f(y) \right) \Leftrightarrow \left(\forall_{x \in P} f(x) = \sum_{x \leq y} \mu(x, y) g(y) \right) \end{matrix}$$
3. Geometric Lattices

a. Definition

i. Definition

A lattices $(L, \leq)$ which is atomic and semimodular

ii. Atomic

Every $x \in L$ is join of finitely many atoms

iii. Semi-modular

If a, b covers c , then $a \vee b$ covers both a and b

b. Example

i. Inclusion

$$L = \left\{ \begin{matrix} \emptyset \\ \{1\}, \{2\}, \{3\} \\ \{1,2\}, \{2,3\}, \{1,3\} \\ \{1,2,3\} \end{matrix} \right\}, \quad \forall_{A \leq B \in L} \mu(A, B) := (-1)^{|B \setminus A|}$$

ii. Divisibility

$$L = \left\{ \begin{matrix} 1 \\ 2,3,5 \\ 6,10,15 \\ 30 \end{matrix} \right\}, \quad \forall_{a,b \in \mathbb{N}^*, n = \frac{a}{b} \in \mathbb{N}^*} \mu(a, b) = \mu(n) = \begin{cases} 1 & n = 1 \\ 0 & n \text{ is not squarefree} \\ (-1)^r & n = p_1 \cdots p_r \end{cases}$$

4. Combinatorial Geometry

a. Definition

i. Notation

$(X, F), \quad F \subseteq \{\text{subset of } X\}$

ii. Property 1: Poset Flat

F include no infinites chains (thus a poset with $\subseteq$)

iii. Property 2:

$$\forall_{A, B \in F} A \cap B \in F$$

iv. Property 3: $0_L, 1_L$ and Atom

$\emptyset, X, \{x\} \in F, \quad \forall x \in X$

v. Property 4: Partition

$\{B \setminus A | B \in F, B \text{ covers } A\}$ gives a partition of $X \setminus A$

b. s

5. '

	Geometry Lattice $(L, < p =)$	Combinatorial Geometry (X, F)	Matroid $M = (E, I)$
"Exchange Axiom"		For $A \subseteq X$ and $x, y \in X$, if $x \notin A$ but $x \in A \cup \{y\}$, then $y \in A \cup \{x\}$	
Independency		A subset $S \subseteq X$ is independent if $\forall x \in S, x \notin S \setminus \{x\}$ (in other words, $\forall x \in S$ there is no flat $A \in F$ such that $S \setminus \{x\} \subseteq A$ but $x \notin A$)	
Span		$\hat{A} := \bigcap_{A \subseteq B \in F} B$	
Isomorphism	A bijective map $\varphi: (L_1, \leq_1) \rightarrow (L_2, \leq_2)$ such that $\forall a, b \in L, a \leq_1 b$ iff $\varphi(a) \leq_2 \varphi(b)$	$\varphi: X_1 \rightarrow X_2$ such that for every $A \subseteq X_1$, we have $A \in F_1$ iff $\varphi(A) \in F_2$	
Basis		Maximal independent subset $S \subseteq A$	
Definition	1) Atomic Every $x \in L$ is join of finitely many atoms 2) Semi-modular If a, b covers c , then $a \vee b$ covers both a and b	1) Property 1: Poset Flat F include no infinites chains (thus a poset with $\subseteq$) 2) Property 2: $\forall_{A, B \in F} A \cap B \in F$ 3) Property 3: $0_L, 1_L$ and Atom $\emptyset, X, \{x\} \in F, \quad \forall x \in X$ 4) Property 4: Partition $\{B \setminus A B \in F, B \text{ covers } A\}$ gives a partition of $X \setminus A$	

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